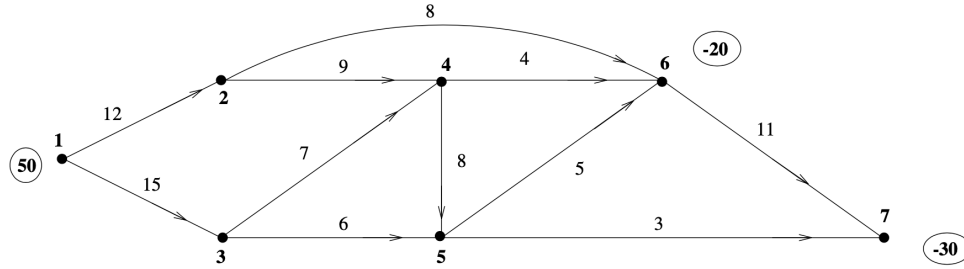


My project will be mainly about Network Simplex and if there is time I'd like to explore the applications in semantic image segmentation. As an introduction I'll describe one of the main network optimization problems, namely the *minimum cost network flow problem*. Suppose you have a directed graph (network) where each vertex is assigned a value called flow and the weights on the edges represent the cost of moving one unit of flow. Consider the example below,

Figure 1: Graph $G = \{E, V\}$ with flow b and weights c



If a vertex has positive flow we call it a source, and if vertex has negative flow we call it a sink. We define Supply, S as

$$S = \sum_{b_i > 0 \in b} b_i,$$

and Demand, D as

$$D = \sum_{b_i < 0 \in b} b_i.$$

We call a network balanced if $S - D = 0$, and an unbalanced network can be made balanced by adding an artificial vertex with flow $S - D$ by a directed edge with zero cost. The goal of the minimum cost network flow problem is to satisfy the supply and demand of each vertex while minimizing the cost. The objective function will look like this,

$$\min z = c^T x$$

where $x_{i,j} \in x$ represents the amount of flow through edge (i, j) . A feasible solution to this problem is an x that satisfies the supply and demand of each vertex. With that context at each vertex i our x must satisfy,

$$\sum_j x_{i,j} - \sum_k x_{k,i} = b_i.$$

One can think of the first sum being flow out of vertex i and the second sum being flow into vertex i . As an example consider vertex 1 and 2 in Figure 1, we get the following constraints

$$x_{1,2} + x_{1,3} = 50,$$

$$x_{2,4} + x_{2,6} - x_{1,2} = 0.$$

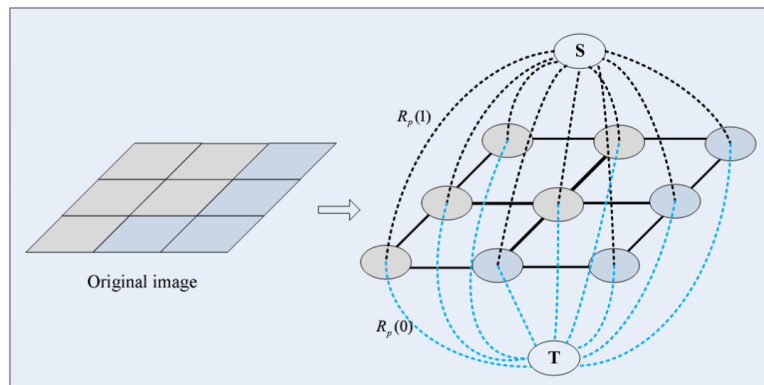
This is done for each vertex i , generating our constraint matrix A which will be size $(\# \text{ of vertices}) \times (\# \text{ of edges})$. Usually there will be some added constraints L and U on the amount of flow that can go through each edge, so most problems will be of the form,

$$\begin{aligned} \text{minimize } z &= c^T x \\ Ax &= b \\ L &\leq x \leq U \end{aligned}$$

Since A is a matrix full of either 1, 0 or -1 and is usually relatively sparse with b also being a sparse vector, the possibility of degeneracy (where x_b in Simplex becomes zero) is common if this LP problem is handed to a general LP solver. Network Simplex modifies the Simplex algorithm to take advantage of the idea that a feasible solution to this problem represents a spanning tree, resulting in a considerable speed up[1](8.4) and resolves common degeneracy and stalling issues by using graph analog of perturbation method[1](8.5).

Following the discussion of Network Simplex, I would like to discuss duality in the context of network algorithms. The *maximum flow problem* which involves finding the maximum flow that can be transmitted between two vertices (a single source and sink), is equivalent to a special form of the minimum cost network flow problem[1](p.278). The dual of this problem is the *minimum cut problem*, which finds the set of edges which must be removed from a graph to separate two vertices into disconnected subgraphs, while minimizing the weight of the removed edges (one of those 'A famous theorem states' in our textbook).

This then would lead to the application in semantic image segmentation as a form of the minimum cut problem. In this context we need to be able to turn an image into a graph, whose edges are appropriately weighted so that solution to the minimum cut problem generates a good segmentation. The following image[2] attempts to describe the this graph is generated,



A user is meant to supply a seed (source and sink pixels s and t), the weights are determined by some similarity function between pixels and the minimum cut problem is solved.

References

- [1] I. Griva, S. Nash, & A. Sofer (2009). *Linear and Nonlinear Optimization*, 2nd ed., SIAM Press.
- [2] Yi, F., & Moon, I. (2012). Image segmentation: A survey of graph-cut methods. 2012 International Conference on Systems and Informatics (ICSAI2012), 1936–1941. <https://doi.org/10.1109/ICSAI.2012.6223428>